

Notes: Syverson (JPE, 2004)
Market Structure and Productivity: A Concrete Example

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1 Big picture

- **Question.** Why do large and persistent productivity differences survive even within narrowly defined industries? The paper asks whether product-market structure, not only technology or management, helps determine which producers survive and how dispersed productivity is within a market.
- **Core mechanism.** If buyers can easily substitute across suppliers, low-productivity plants have less room to survive. Greater substitutability intensifies competition, truncates the productivity distribution from below, raises minimum and average productivity, lowers dispersion, and increases average plant size.
- **Concrete setting.** The empirical setting is U.S. ready-mixed concrete, SIC 3273. This industry is useful because concrete is physically homogeneous relative to many manufactured products, but transport costs are very high. Those transport costs make markets local and make spatial substitutability central.
- **Source of variation.** The paper exploits differences in local demand density. A dense market requires more concrete plants per unit area. When plants are physically closer together, customers can more easily switch among suppliers, so spatial substitutability rises.
- **Main prediction.** Dense concrete markets should have: (i) less productivity dispersion, (ii) higher median and output-weighted average TFP, (iii) a higher lower tail of TFP, and (iv) larger average plants.
- **Empirical bottom line.** The data support all of the main predictions. In the benchmark regressions, demand density is negatively related to the interquartile range of TFP and positively related to median TFP, output-weighted TFP, tenth-percentile TFP, and average plant output.
- **Economic takeaway.** Market structure helps shape observed productivity distributions. In denser markets, competition removes the least productive producers; this is a selection effect, not necessarily a within-plant technology improvement.

2 Incremental contribution of the paper

- **From technology-only explanations to demand-side selection.** Earlier plant-level productivity work documented persistent heterogeneity and often emphasized management, capital vintage, R&D, or learning. This paper shows that product-market substitutability can also sustain or compress productivity dispersion.
- **A market-structure theory of productivity dispersion.** The paper provides a simple spatial model in which the lower bound of the productivity distribution is endogenous and depends on demand density.
- **A concrete industry case study.** By focusing on one four-digit SIC industry, the paper greatly reduces variation in technology across producers and isolates market-side forces.
- **Use of physical output.** The paper measures concrete output in cubic yards rather than only deflated revenue. This is important because revenue-based productivity can confound efficiency with local price variation.

- **Geographic identification.** High transport costs turn the national ready-mixed concrete industry into many quasi-independent local markets. This makes cross-market differences in demand density empirically informative.
- **Selection-based agglomeration.** The paper offers an agglomeration effect that does not rely on Marshallian externalities, information spillovers, or internal increasing returns. Productivity is higher in dense markets because low-productivity producers are selected out.
- **Bridge to competition and productivity.** The findings support the broader idea that competition improves efficiency, but the channel is between-plant selection rather than necessarily within-plant productivity upgrading.

3 Data and measurement

3.1 Industry and plant data

- **Industry.** The paper studies ready-mixed concrete, SIC 3273.
- **Years.** The main plant-level data come from the 1982, 1987, and 1992 Census of Manufactures.
- **Plant coverage.** Each Census of Manufactures year contains information on roughly 5,200 ready-mixed concrete plants.
- **Administrative records.** Very small plants often have imputed production variables. These administrative record plants are excluded from the main productivity sample, although their locations can still be used when counting market producers.
- **Product-level advantage.** The paper augments plant records with census product data. Ready-mixed concrete is unusual because physical output in cubic yards is available for almost all non-imputed plants.
- **Output measure.** The preferred output measure is cubic yards of concrete shipped. This avoids treating high local prices as high productivity.

3.2 Productivity measurement

The benchmark productivity measure is a standard total factor productivity index:

$$TFP_{it} = q_{it} - a_l l_{it} - a_k k_{it} - a_m m_{it} - a_e e_{it}, \quad (1)$$

where q_{it} is log physical output, and labor, capital, materials, and energy inputs are weighted by industry-level input cost shares.

- **Labor.** Production worker hours are observed directly. Nonproduction worker hours are imputed using the average annual hours of nonproduction employees in the corresponding two-digit industry.
- **Capital.** Equipment and building capital stocks are reported book values converted to real values using industry-level ratios. Rented capital is capitalized using rental rates.
- **Materials and energy.** Materials and energy expenditures are deflated with industry-level deflators.
- **Constant returns benchmark.** The index assumes constant returns to scale. Robustness checks consider scale elasticities of 0.9 and 1.1 and also estimate a production function directly.

3.3 Market definition

- **Local markets.** Markets are Bureau of Economic Analysis component economic areas (CEAs), which are groups of economically connected contiguous counties.
- **Why CEAs fit concrete.** Ready-mixed concrete has high transport costs and must be delivered quickly. The industry therefore consists of local markets rather than a single national market.
- **Number of markets.** The CEA system creates 348 mutually exclusive and exhaustive U.S. markets.
- **Main estimation sample.** The benchmark market-year sample includes CEA-years with at least five non-imputed plants with TFP data. This leaves 665 market-year observations.

3.4 Demand density

- **Main variable.** Demand density is measured as

$$\log \left(\frac{\text{construction sector employees in CEA-year}}{\text{square miles in CEA}} \right). \quad (2)$$

- **Why construction employment.** The construction sector buys almost all ready-mixed concrete output, so local construction employment is a natural demand shifter.
- **Why plausibly exogenous.** Ready-mixed concrete is only a small share of construction costs, so changes in concrete market competitiveness are unlikely to generate large reverse effects on local construction employment.
- **Interpretation.** Demand density is not just market size. Holding total demand fixed, higher density means plants and buyers are closer together, making substitution easier.

3.5 Dependent variables

The empirical tests use market-level moments of the productivity and size distributions:

- **TFP dispersion.** Interquartile range of plant-level TFP in the market-year.
- **Median TFP.** Central tendency robust to outliers.
- **Output-weighted average TFP.** Captures both plant productivity and reallocation of output toward more productive plants.
- **Tenth-percentile TFP.** Proxy for the market lower bound of productivity.
- **Producer-demand ratio.** Number of market producers divided by construction employment.
- **Average plant output.** Average log cubic yards shipped by plants in the market.

4 Model and empirical predictions

4.1 Spatial demand setup

- Consumers are uniformly located around a circle of unit circumference.
- Demand density is D consumers per unit length.

- Each consumer buys one unit if the delivered price is below the reservation value.
- Delivered price equals factory-door price plus transport cost:

$$p^c = p + tx, \quad (3)$$

where x is distance and t captures transport costs.

- A higher D does not by itself change the physical distance around the circle, but it supports more plants per unit area. More plants imply shorter distances among suppliers and more substitution possibilities.

4.2 Entry and production

- Potential entrants first pay a sunk setup cost s to learn their own marginal cost c_i .
- Cost draws come from a common distribution $g(c)$ on $[0, c_u]$.
- After learning c_i , entrants decide whether to produce. Production requires a fixed cost f .
- Active producers are placed evenly around the circle and set factory-door prices.
- Producers know their own costs, but not rivals' realized costs at the time of entry/production decisions. This asymmetric information makes the equilibrium analytically tractable.

4.3 Pricing and expected profit

For neighboring plants i and j , the indifferent consumer satisfies

$$p_i + tx_{ij} = p_j + t \left(\frac{1}{n} - x_{ij} \right), \quad (4)$$

where n is the number of active producers.

A producer's expected profit can be written as

$$E(\pi_i) = \left[\frac{E(p) - p_i}{t} + E \left(\frac{1}{n} \right) \right] (p_i - c_i) D - f. \quad (5)$$

The optimal price is

$$p_i = \frac{1}{2}c_i + \frac{1}{2}E(p) + \frac{t}{2}E \left(\frac{1}{n} \right). \quad (6)$$

The expected market price among rivals is

$$E(p) = E(c) + tE \left(\frac{1}{n} \right). \quad (7)$$

The key comparative statics follow from the fact that markups, expected shares, and expected profits are decreasing in c_i . Low-cost producers can profitably operate under tougher competition; high-cost producers cannot.

4.4 Cutoff productivity and truncation

A marginal producer with cutoff cost c^* has zero expected operating profit. The cutoff is characterized by

$$c^* = E(c) + 2tE\left(\frac{1}{n}\right) - \sqrt{\frac{4tf}{D}}. \quad (8)$$

The equilibrium cost distribution among producers is the ex ante cost distribution truncated above at c^* :

$$E(c) = \int_0^{c^*} \frac{cg(c)}{G(c^*)} dc. \quad (9)$$

The expected value of entry is driven to zero by free entry:

$$V^e = \int_0^{c^*} \left[\frac{D}{4t} \left(c^* - c + \sqrt{\frac{4tf}{D}} \right)^2 - f \right] g(c) dc - s = 0. \quad (10)$$

The model implies

$$\frac{dc^*}{dD} < 0. \quad (11)$$

Interpretation: as demand density rises, entry becomes more attractive. More entry puts producers closer together, which raises substitutability and competitive pressure. The highest-cost producers can no longer profitably survive. Thus, higher density lowers the cost cutoff and raises the productivity lower bound.

4.5 Empirical predictions

Because productivity is inversely related to marginal cost, a lower cost cutoff c^* implies:

1. **Lower productivity dispersion.** The low-productivity tail is truncated, so the interquartile range should fall.
2. **Higher median productivity.** Removing low-productivity plants shifts the distribution upward.
3. **Higher output-weighted productivity.** More efficient plants survive and absorb more market share.
4. **Higher lower-bound productivity.** The tenth-percentile TFP should be higher in denser markets.
5. **Larger average plants.** Free entry plus lower markups imply that surviving producers in denser markets must sell more output on average to cover setup and fixed costs.

5 Identification and empirical design

5.1 Regression specification

The main empirical specification is

$$y_{it} = \beta_0 + \beta_d \text{dens}_{it} + X'_{it} B + \epsilon_{it}, \quad (12)$$

where y_{it} is a market-level moment in CEA i and year t , and dens_{it} is log construction employment per square mile.

5.2 Predicted signs

Outcome	Prediction for demand density	Economic reason
TFP interquartile range	Negative	Low-productivity tail is truncated
Median TFP	Positive	Productivity distribution shifts up
Output-weighted TFP	Positive	Output reallocates to more productive plants
Tenth-percentile TFP	Positive	Lower bound of productivity rises
Producer-demand ratio	Negative	More demand per producer in dense markets
Average plant output	Positive	Surviving plants are larger

5.3 Control variables

The richest benchmark model includes year fixed effects and local demand controls:

- demographics: race, age structure, education, marriages per population;
- housing and income variables: owner-occupied housing, median home value, median income;
- local transportation/urban features: fraction of households with at least two automobiles;
- industry structure: output-weighted specialization in ready-mixed concrete;
- short-run market conditions: five-year growth in local construction employment.

A further specification adds an overall employment-density measure in the spirit of Ciccone and Hall to test whether the concrete-specific demand-density result is distinct from general urban density effects.

5.4 Key identification logic

- **Within-industry focus.** By studying only ready-mixed concrete, the paper reduces technological heterogeneity across plants.
- **Physical output.** Cubic yards reduce the risk that price differences are misread as productivity differences.
- **Local market segmentation.** High transport costs make CEAs plausible local markets.
- **Demand density as spatial substitutability.** Higher construction employment per square mile implies denser plant networks and easier switching among suppliers.
- **Exogeneity argument.** Construction demand drives concrete demand, but ready-mixed concrete is too small a share of construction costs for concrete competition to plausibly determine construction density.

5.5 Main threats and robustness checks

- **Small-market measurement.** Market moments may be noisy with few plants. The paper changes the minimum plant cutoff from five to two and ten.
- **Capital imputation.** Small plants often have imputed capital. The paper checks labor productivity as an alternative to TFP.

- **Revenue versus physical output.** The paper checks revenue-based TFP; differences help reveal the role of local prices.
- **Returns to scale.** The paper recomputes TFP under scale elasticities of 0.9 and 1.1.
- **Production function estimation.** The paper estimates a production function using local construction demand instruments and finds a scale elasticity near one.
- **Long-run versus short-run variation.** The paper averages TFP and density over time to create a single cross section.
- **Density versus size.** The paper adds market size to show that density, not only total demand, matters.

6 Figures and tables

6.1 Figure 1 and Figure 2: Initial distributional evidence

Figure 1 read:

- Figure 1 plots kernel densities of plant TFP for plants in markets above and below median demand density.
- The high-density distribution places less mass in the low-productivity left tail and more mass around higher productivity values.
- The figure visually supports the model’s truncation prediction: dense markets look like they have selected away some low-productivity producers.

Figure 2 read:

- Figure 2 plots kernel densities of logged output for plants in markets above and below median demand density.
- The high-density distribution is shifted to the right, showing that plants in dense markets are larger.
- This supports the model’s prediction that denser markets have larger surviving producers, even without internal scale economies.

Economic message: The two figures jointly show the central empirical pattern before any regression: dense markets have both more productive and larger plants.

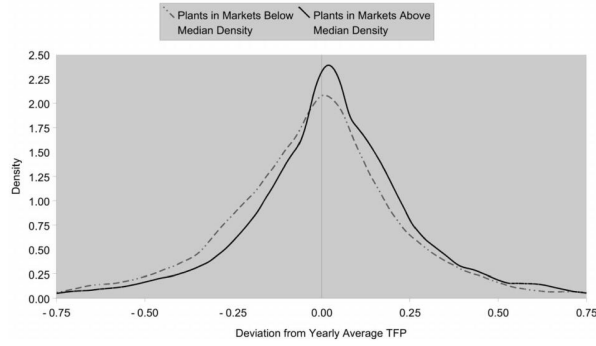


FIG. 1.—TFP kernel density estimates, plants in markets above and below median demand density

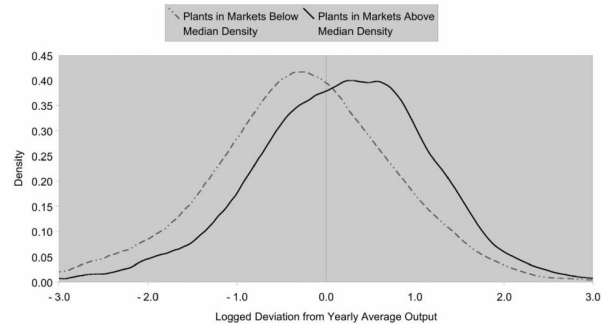


FIG. 2.—Output (logged shipments in cubic yards) kernel density estimates, plants in markets above and below median density

(a) Figure 1: TFP density by market demand density.

High-density markets have less low-productivity mass.

(b) Figure 2: Output density by market demand density.

High-density markets have larger plants.

6.2 Figure 3 and Table A1: Simulation and production-function support

Figure 3 read:

- Figure 3 shows a simulation allowing ex post entrants to observe incumbents' costs and choose the best location.
- Even in this less restrictive case, the upper-bound marginal cost of a potential entrant falls with demand density.
- Average units sold per producer rise with demand density.
- The figure supports the robustness of the model's comparative statics: higher density increases selection pressure and raises average plant size.

Table A1 read:

- Table A1 reports production-function estimates using local construction activity as instruments.
- The first-stage F-statistic for the demand instruments is 35.2, indicating strong relevance.
- The sum of demand instrument coefficients is positive, consistent with local construction demand increasing concrete input use.
- The estimated scale elasticity is 0.996 with standard error 0.029, essentially constant returns to scale.

Economic message: Figure 3 supports the model mechanism under a more permissive entry scenario, and Table A1 supports the benchmark constant-returns TFP construction.

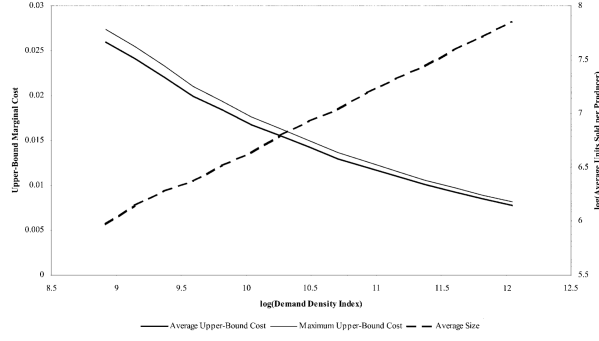


FIG. 3.—Simulated upper-bound marginal costs of ex post entrants and average producer size vs. demand density

FIRST STAGE		SECOND STAGE				
<i>F</i> -Statistic for Demand Instruments	Sum of Demand Instrument Coefficients	δ_{1982}	δ_{1987}	γ_{multi}	γ_s	R^2
35.2 ($p < .001$)	1.025 (.115)	-.115 (.010)	-.079 (.012)	.070 (.010)	.996 (.029)	.902

NOTE.—Reported standard errors (in parentheses) are robust and are clustered by CEA (local market). Production function coefficients: δ_{1982} : 1982 year fixed effect; δ_{1987} : 1987 year fixed effect; γ_{multi} : multiplant firm effect; γ_s : scale elasticity.

(a) Figure 3: Simulated ex post entry and average size.

Selection pressure rises and average size increases with density.

(b) Table A1: Production-function estimates.

The estimated scale elasticity is nearly one.

6.3 Table 1 and Table 2: Descriptive statistics and market structure

Table 1 read:

- Table 1 summarizes productivity moments, size moments, and demand density.
- Across 665 market-year observations, the mean interquartile TFP range is 0.275, implying large within-market productivity dispersion.
- The mean median TFP is 4.557, and mean output-weighted average TFP is 4.599.
- The mean tenth-percentile TFP is 4.169 with a large standard deviation of 0.450, indicating substantial variation in lower-bound productivity across markets.
- The mean log average plant output is 10.637, with a standard deviation of 0.543.
- Demand density also varies substantially across markets, with a standard deviation of 1.439 in log construction employment per square mile.

Table 2 read:

- Table 2 reports the number of producers across all markets and in the estimation sample.
- In all markets with at least one plant, the mean number of plants is 15.0; the median is 10.
- In the estimation sample, the mean number of plants is 20.6 and the median is 14.
- The estimation sample therefore focuses on markets with enough plants to measure distributional moments reliably.

Economic message: The data contain large cross-market variation in density, productivity dispersion, lower-tail productivity, and market size. This variation is what identifies the empirical relationship between market structure and productivity moments.

TABLE 1
DESCRIPTIVE STATISTICS: PRODUCTIVITY AND SIZE MOMENTS AND DEMAND DENSITY

Variable	Mean	Standard Deviation	Skewness	75th–25th Percentile Range	90th–10th Percentile Range
TFP dispersion (inter-quartile range)	.275	.147	1.332	.164	.353
Median TFP	4.557	.122	–.028	.133	.278
Output-weighted average TFP	4.599	.177	1.627	.183	.360
10th-percentile TFP	4.169	.450	–6.114	.293	.611
ln(plants per demand unit)	–6.447	.703	–.078	.978	1.827
ln(average plant output)	10.637	.543	.086	.752	1.343
Demand density: ln(construction employment per square mile)	.495	1.439	.050	1.700	3.691
All 1,044 U.S. markets	.317	1.407	.066	1.534	3.548
exp(demand density)	4.801	11.233	6.009	3.269	10.777
ln(number of construction employees)	9.229	1.035	.401	1.547	2.690
Plant-level TFP (N=10,613)	4.559	.354	–1.655	.277	.637

NOTE.—This table shows descriptive statistics for key variables. Figures are measured across 665 market-year observations unless otherwise stated. See the text for details.

(a) Table 1: Descriptive statistics.

Large variation exists in productivity moments and demand density.

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TABLE 2
DESCRIPTIVE STATISTICS: NUMBER OF PRODUCERS BY MARKET

Variable	Mean	Minimum	25th Percentile	Median	75th Percentile	Maximum
A. All Markets with One or More Plants (N=1,031)						
Number of plants	15.0	1	6	10	18	102
Number with TFP data	10.3	1	3	7	12	79
B. Estimating Sample						
Number of plants	20.6	5	10	14	25	102
Number with TFP data	14.5	5	7	10	17	79

NOTE.—See note to table 1.

(b) Table 2: Number of producers by market.

The benchmark sample uses markets with at least five plants with TFP data.

6.4 Table 3 and Table 4: Main regression results and demand controls

Table 3 read:

- Table 3 is the core empirical table.
- For TFP dispersion, the demand-density coefficient is negative in all specifications and significant. In the model with year effects and demand controls, the coefficient is -0.029.
- Median TFP and output-weighted average TFP are positively related to density in the first three specifications. After adding the overall density control in model 4, those coefficients become smaller and insignificant.
- Tenth-percentile TFP is positively related to density in all specifications. In the richest model, the coefficient is 0.056 and significant.
- Producer-demand ratio is negatively related to density, implying fewer producers per unit of demand in denser markets.
- Average plant output is positively related to density. The richest specification gives a coefficient of 0.142.
- These results directly support the model's selection and size predictions.

Table 4 read:

- Table 4 summarizes which demand controls are significant in the full-control specifications.
- Year effects matter: 1982 and 1987 are significant in several productivity regressions.
- Median income is positively associated with several productivity measures.
- Controls do not remove the core demand-density relationships in Table 3.

Economic message: The main regression evidence is consistent with demand-density-driven selection. The strongest and most robust effects are on productivity dispersion, lower-bound productivity, and plant size.

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TABLE 3
MAIN REGRESSION RESULTS: LOCAL PRODUCTIVITY AND SIZE MOMENTS—DEMAND
DENSITY COEFFICIENTS

Dependent Variable	Regression Statistic	Model 1	Model 2	Model 3	Model 4
TFP dispersion (interquartile range)	R^2	.018	.036	.092	.092
	Demand density coefficient	-.014* (.004)	-.015* (.004)	-.029* (.008)	-.031* (.010)
Median TFP	R^2	.059	.289	.321	.322
	Demand density coefficient	.021* (.003)	.018* (.003)	.012* (.005)	.008 (.006)
Output-weighted average TFP	R^2	.045	.125	.162	.162
	Demand density coefficient	.026* (.004)	.024* (.004)	.016* (.008)	.012 (.010)
10th-percentile TFP	R^2	.033	.033	.058	.059
	Demand density coefficient	.057* (.010)	.056* (.010)	.065* (.019)	.056* (.022)
Producer-demand ratio	R^2	.570	.584	.708	.711
	Demand density coefficient	-.369* (.015)	-.363* (.015)	-.313* (.022)	-.278* (.030)
Average plant output	R^2	.334	.376	.557	.563
	Demand density coefficient	.218* (.012)	.211* (.012)	.184* (.017)	.142* (.023)
Year dummies		no	yes	yes	yes
Demand controls		no	no	yes	yes ¹

NOTE.—The table shows the estimated coefficients on demand density when various moments of the local productivity and size distributions are regressed on demand density and, when applicable, a set of demand controls. Specifications are given by column and dependent variables by row. The sample consists of 665 region-year observations with at least five plants for which I have nonimputed production data. Reported standard errors (in parentheses) are robust to heteroskedasticity.

(a) Table 3: Demand density coefficients.

Dense markets have less dispersion, higher lower-tail TFP, and larger plants.

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TABLE 4
MAIN REGRESSION RESULTS: LOCAL PRODUCTIVITY AND SIZE MOMENTS—SIGNIFICANCE
OF DEMAND CONTROLS

Dependent Variable	Negative and Significant	Positive and Significant
TFP dispersion (interquartile range)	1982, 1987, married, college, auto2, spec	Median income
Median TFP	1982, 1987, nonwhite, occupied	Median income
Output-weighted average TFP	1982, 1987, black	Median income
10th-percentile TFP	None	Married, college
Producer-demand ratio	Married, nonwhite, college, median income, growth	1982, over25, spec
Average plant output	1982, over25	1987, married, college, median house, growth

NOTE.—The table shows, by dependent variable, the significance of the demand controls included in the specification corresponding to col. 3 in table 3. All demand controls are included in each regression, so those not reported were statistically insignificant. Key to demand controls: 1982: year dummy; 1987: year dummy; married: fraction of population that is married; nonwhite: fraction of population that is nonwhite; college: fraction of population with a college education; over25: fraction of population over 25 years old; median income: logged median household income; auto2: fraction of households with at least two cars; occupied: fraction of owner-occupied housing units; median house: logged value of median house; growth: demand growth over past five years (log change in construction sector employees); spec: owner-occupied average revenue share of real-estate concrete among concrete plants.

(b) Table 4: Significant demand controls.

Controls shift some estimates but do not overturn the main mechanism.

6.5 Table 5 and Table 6: Robustness checks

Table 5 read:

- Table 5 repeats robustness checks in a bivariate specification with only demand density and a constant.
- Across alternative samples, labor productivity, revenue-based TFP, scale assumptions, estimated TFP, averaged cross sections, and market-size controls, the demand-density coefficients generally keep the expected signs.
- TFP dispersion remains negatively related to demand density across all specifications.
- Tenth-percentile TFP remains positively related to demand density across all specifications.
- Average plant output remains positively related to demand density where reported.

Table 6 read:

- Table 6 repeats robustness checks with year effects and demand controls.
- The dispersion and tenth-percentile TFP results are especially robust.
- The median and output-weighted average TFP effects are more sensitive, particularly when using revenue-based TFP or allowing strong increasing returns.
- The average plant output effect remains positive and significant in all reported specifications.

Economic message: Robustness checks reinforce the strongest part of the paper: density compresses productivity dispersion from below and raises producer size. The average-TFP result is supportive but less robust once alternative density or revenue-based measures are introduced.

TABLE 5
ROBUSTNESS CHECKS—BIVARIATE SPECIFICATION (Only Demand Density and a Constant Included)

DEPENDENT VARIABLE		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
SPECIFICATION											
TFP dispersion	R^2	.018	.026	.016	.018	.020	.015	.011	.019	.018	.019
	Demand density	-.014*	-.020*	-.011*	-.016*	-.012*	-.012*	-.011*	-.014*	-.014*	-.016*
		(.004)	(.006)	(.005)	(.008)	(.003)	(.004)	(.004)	(.004)	(.005)	(.006)
Median TFP	R^2	.059	.056	.021	.359	.034	.167	.003	.050	.074	.072
	Demand density	.021*	.021*	.011*	.072*	.013*	.040*	.005	.017*	.021*	.015*
		(.003)	(.003)	(.004)	(.007)	(.003)	(.003)	(.003)	(.003)	(.004)	(.004)
Quantity-weighted average TFP	R^2	.045	.038	.009	.413	.028	.113	.004	.034	.037	.036
	Demand density	.020*	.027*	.012*	.080*	.014*	.044*	.008	.022*	.021*	.016*
		(.004)	(.004)	(.005)	(.006)	(.003)	(.004)	(.004)	(.004)	(.005)	(.006)
10th-percentile TFP	R^2	.033	.024	.028	.142	.054	.032	.015	.028	.071	.040
	Demand density	.057*	.046*	.027*	.120*	.027*	.070*	.038*	.053*	.064*	.032*
		(.010)	(.008)	(.008)	(.013)	(.004)	(.009)	(.010)	(.010)	(.014)	(.013)
Producer-demand ratio	R^2	.579	.510	.576					.207	.713	
	Demand density	-.369*	-.361*	-.365*					-.320*	-.212*	
		(.015)	(.012)	(.023)					(.033)	(.016)	
Average plant output	R^2	.334	.269	.311		.353		.222	.467		
	Demand density	.218*	.211*	.206*		.227*		.182*	.098*		
		(.012)	(.011)	(.017)		(.013)		(.020)	(.015)		

Notes.—This table shows results from robustness tests on the results presented in table 3. Heteroskedasticity-robust standard errors are in parentheses. Average plant output is revenue-based in specification 5. Key to specifications: (1) benchmark (from table 3); (2) the minimum number of plants in the market cell with TFP data is two rather than five ($N=974$); (3) the minimum number of plants in the market cell with TFP data is 10 rather than five ($N=540$); (4) labor productivity (logged output per worker hour) instead of TFP market capitalizer controls; (5) revenue-based TFP measure; (6) assumed scale elasticity of 0.9; (7) assumed scale elasticity of 1.1; (8) estimated TFP from the production function; (9) a single cross section using average TFP and density measures ($N=302$); (10) a market size control is included.

* Significant at the 5 percent level.

TABLE 6
ROBUSTNESS CHECKS (Year Effects and Demand Controls Included)

DEPENDENT VARIABLE		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
SPECIFICATION											
TFP dispersion	R^2	.092	.065	.103	.044	.083	.062	.088	.093	.072	.090
	Demand density	-.020*	-.034*	-.008	-.031*	-.021*	-.024*	-.028*	-.033*	-.017*	-.020*
		(.008)	(.009)	(.008)	(.012)	(.005)	(.008)	(.009)	(.009)	(.007)	(.009)
Median TFP	R^2	.321	.181	.429	.414	.386	.393	.252	.110	.131	.337
	Demand density	.012*	.017*	.006	.054*	.002	.028*	.000	.009	.019*	.010*
		(.005)	(.005)	(.007)	(.011)	(.004)	(.005)	(.006)	(.005)	(.005)	(.005)
Quantity-weighted average TFP	R^2	.162	.133	.165	.469	.311	.249	.120	.078	.107	.168
	Demand density	.016*	.023*	.013	.056*	-.004	.032*	-.000	.014	.016*	.011
		(.008)	(.006)	(.011)	(.009)	(.005)	(.008)	(.008)	(.008)	(.006)	(.008)
10th-percentile TFP	R^2	.038	.021	.110	.176	.154	.076	.041	.066	.121	.063
	Demand density	.065*	.058*	.033*	.116*	.019*	.079*	.048*	.062*	.063*	.049*
		(.019)	(.013)	(.014)	(.020)	(.007)	(.017)	(.021)	(.019)	(.017)	(.020)
Producer-demand ratio	R^2	.706	.630	.779					.237	.760	
	Demand density	-.313*	-.324*	-.294*					-.361*	-.242*	
		(.022)	(.018)	(.025)					(.037)	(.019)	
Average plant output	R^2	.557	.478	.633		.609			.487	.610	
	Demand density	.184*	.199*	.163*		.200*			.107*	.129*	
		(.017)	(.016)	(.026)		(.017)			(.022)	(.017)	

Notes.—This table shows results from robustness tests on the results presented in table 3. Heteroskedasticity-robust standard errors are in parentheses. Average plant output is revenue-based in specification 5. Key to specifications: (1) benchmark (from table 3); (2) the minimum number of plants in the market cell with TFP data is two rather than five ($N=974$); (3) the minimum number of plants in the market cell with TFP data is 10 rather than five ($N=540$); (4) labor productivity (logged output per worker hour) instead of TFP market capitalizer controls; (5) revenue-based TFP measure; (6) assumed scale elasticity of 0.9; (7) assumed scale elasticity of 1.1; (8) estimated TFP from the production function; (9) a single cross section using average TFP and density measures ($N=302$); (10) a market size control is included.

* Significant at the 5 percent level.

(a) Table 5: Robustness, bivariate models.

The signs are broadly stable across alternative measurement choices.

(b) Table 6: Robustness with controls.

Dispersion, lower-tail productivity, and size results remain the most robust.

7 Short summary

- The paper explains persistent within-industry productivity dispersion through a demand-side selection channel.
- When producers are spatially close together, consumers can switch suppliers more easily.
- Greater substitutability intensifies competition and makes survival harder for low-productivity plants.
- The model predicts that denser markets should have less productivity dispersion, higher minimum productivity, higher average productivity, and larger surviving producers.
- Ready-mixed concrete provides a clean setting because output is relatively homogeneous, transport costs are high, and physical output is observed.
- Demand density is measured as construction employment per square mile in a local economic area.
- The empirical evidence supports the model: dense markets have lower TFP dispersion, higher lower-tail TFP, higher average TFP, lower producer-demand ratios, and larger plants.
- Robustness checks show that the dispersion, lower-bound, and plant-size results are especially stable.
- The findings imply that competition can raise productivity through selection, not only through within-plant improvements.

8 Conclusion

8.1 Core conclusion

Syverson shows that market structure affects the equilibrium distribution of producer productivity. In ready-mixed concrete, markets with high demand density have more spatially substitutable producers. This increased substitutability strengthens competition and eliminates the least productive

plants. As a result, dense markets have higher lower-tail productivity, less productivity dispersion, and larger average plants.

8.2 Economic intuition

The intuition is simple but powerful. If consumers have few nearby alternatives, a low-productivity plant can still keep customers because switching suppliers is costly. If consumers have many nearby alternatives, the same inefficient plant cannot keep enough customers at profitable prices. Demand density therefore acts like a competitive selection force.

8.3 Incremental contribution revisited

The paper contributes by moving the productivity literature beyond plant technology alone. It shows that the demand side - specifically the degree of product substitutability - can determine how much productivity heterogeneity survives in equilibrium. The paper also adds a selection-based agglomeration channel: producers are more productive in dense markets not necessarily because of knowledge spillovers or local externalities, but because weaker plants are selected out.

8.4 Implications

- **For productivity research.** Observed productivity dispersion is partly a market-structure outcome, not only a technology outcome.
- **For competition policy.** Increased competition may raise aggregate productivity by reallocating market share away from less productive producers.
- **For urban economics.** Dense markets may have higher productivity through selection, even without agglomeration externalities.
- **For trade and transport infrastructure.** Lower transport costs can increase effective substitutability and therefore sharpen productivity selection.

8.5 Limitations

- The study is a detailed case study of one industry; external validity to other industries depends on whether substitutability can be measured and whether product markets are similarly segmented.
- Demand density explains only a modest part of cross-market variation in productivity moments, so supply-side factors and unmeasured product differentiation remain important.
- The model abstracts from dynamic adjustment, endogenous technology upgrading, and detailed multiproduct or multiplant firm behavior.
- The average-productivity effect is less robust than the dispersion, lower-tail, and size effects.

8.6 Takeaway

The paper's main lesson is that productivity differences are not only about how plants produce; they are also about the markets in which plants compete. In dense, highly substitutable markets, competitive selection raises the productivity floor and compresses the productivity distribution.